

Christopher Ortega

LLM Infrastructure • LLMOps • Systems Architecture

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SUMMARY

Systems architect with 9 years across backend services, streaming data platforms, and bare-metal infrastructure — building with LLMs since 2023. I serve 27B+ parameter models on local hardware (GPU telemetry, KV-cache management, OOM-proof context sizing), build retrieval that returns the right context with citations, and ship evaluation gates that block bad releases. HackAPrompt 2025 Top-10.

HIGHLIGHTS

- | | |
|---------------|------------------|
| — Top-10 (#7) | — 27B+ params |
| — 10+ TB/day | — RBAC + OpenAPI |

EXPERIENCE

LLM Infrastructure Consultant & Systems Architect — Effusion Labs · Westminster, CO Dec 2023

- Built fault-tolerant bare-metal LLM serving for 27B+ parameter models — tokens/sec optimization with dynamic KV-cache management driven by NVIDIA SMI telemetry to prevent OOM.
- Engineered automated memory-profiling tools that calculate hardware-safe context windows, eliminating runtime memory failures and hangs.
- HackAPrompt 2025 — Top-10 (7th): turned jailbreak tactics into repeatable tests and mitigations; authored evaluation suites (200+ prompts) with Positive/Negative/Risk cases covering groundedness (“no citation → no claim”), refusal-appropriateness, unsafe-rate, and p50/p95; deltas: +9 accuracy points, p95 ≤1.8s, unsafe probes ~0 on re-tests.
- Patent Intelligence Agent (consulting, private client): constrained Q&A over patent corpora with hybrid semantic + CPC/IPC filtering, claim/abstract parsing, citation-trail review, and landscape digests; outputs include inline citations and limitation mapping; added RBAC, prompt-safety controls, and clear notices.
- Built resilient process orchestration governing AI server lifecycles — mitigating zombie processes, hardware timeouts, and silent failures — with persistent file-based state keeping continuous context across isolated multi-agent sessions.
- Local LLM workflows with modular orchestration (prompt-rewrite steps, LoRA adapters, graph-style configs) to compare prompts/models offline with reproducible inputs.

RAG · Guardrails · GPU telemetry · KV cache · Quantization · RBAC · OpenAPI · Elasticsearch · LangChain · PyTorch · Ollama · Rust · Linux internals

Senior Software Engineer / Platform Tech Lead — Charter Communications · Denver, CO Feb 2019 — Nov 2023

- Shaped backend architecture for segmentation/measurement across millions of subscribers; aligned product, SRE, and engineering on release criteria and operational readiness.
- Implemented role-based access with Spring Security + JWT; enforced OpenAPI hygiene and versioning to keep partner integrations stable across releases.
- Built Scala/Spark batch + stream services processing 10+ TB/day; added Kafka·Kinesis·Lambda pathways to support near-real-time activation triggers.
- Drove Elasticsearch indexing/aggregation patterns for complex eligibility queries; containerized on Kubernetes and enforced CI/CD checks in GitLab.

Scala · Spark · Kafka · Kinesis · AWS EMR · Lambda · Elasticsearch · Kubernetes · GitLab CI · Spring Security · JWT · OpenAPI

- Moved legacy interfaces to React + TypeScript; introduced shared components and patterns adopted across internal tools.
- Delivered accessible dashboards tied to real-time targeting APIs; partnered with backend and QA on staged rollouts.

React · TypeScript · Design systems · Accessibility · Dashboards · API integration

SELECTED PROJECTS

HackAPrompt Top-10 (7th) — Jailbreak and prompt-injection challenge (2025)

[link](#)

Placed 7th; built reproducible jailbreak tests with shadow and canary rollouts that enforced accuracy and safety thresholds.

Patent Intelligence Agent — Agentic Q&A over patent corpora

[link](#)

Hybrid search with CPC/IPC filters and citation trails; maps claims and limitations and returns answers with inline citations and scope notes.

ComfyUI-TTools — Open-source SD3 resolution utilities

[link](#)

Python/JSON tooling for Stable Diffusion 3 resolution optimization.

LLM Release Watcher — Safe, measurable LLM releases

[link](#)

Monitors accuracy, latency, and safety metrics with shadow runs, timed canaries, and automatic rollback.

RAG Quality Kit — Retrieval reliability

Small gold sets and sanity checks for chunking, indexing, and retrieval; scripts to catch regressions early and visualize coverage.

Local-LLM Inference Graphs — Reproducible offline A/B

Configurable offline A/B harness for prompts, models, and tools with LoRA adapters and rewrite steps; designed for reproducible experiments.

SKILLS & TOOLS

AI / LLM: Prompt design, Retrieval / RAG, Evaluation / metrics, Safety controls, GPU resource allocation, Model quantization, PyTorch, Transformers, LangChain, Ollama

Backend & Data: Python, Rust, Scala (background), Spark, Elasticsearch, Kafka, Kinesis, pandas, NumPy

Cloud & DevOps: AWS, Kubernetes, Terraform, Docker, GitLab CI/CD, Linux internals

APIs & Practices: REST, OpenAPI, JWT/OAuth2, Release checks, Observability, Response playbooks, TRDs, BDD acceptance

Tools: VS Code, Git, GitLab, OpenAPI Generator, Postman, cURL, jq, DBeaver, nvidia-smi, Markdown / MDX, Eleventy (11ty), Tailwind 4, Nunjucks

EDUCATION

B.S., Computer Information Systems — Metropolitan State University of Denver

A.S., Business Management — Front Range Community College